



Advancing Services for Irish Education, Research & Academia

Lessons from the trenches operating WFO

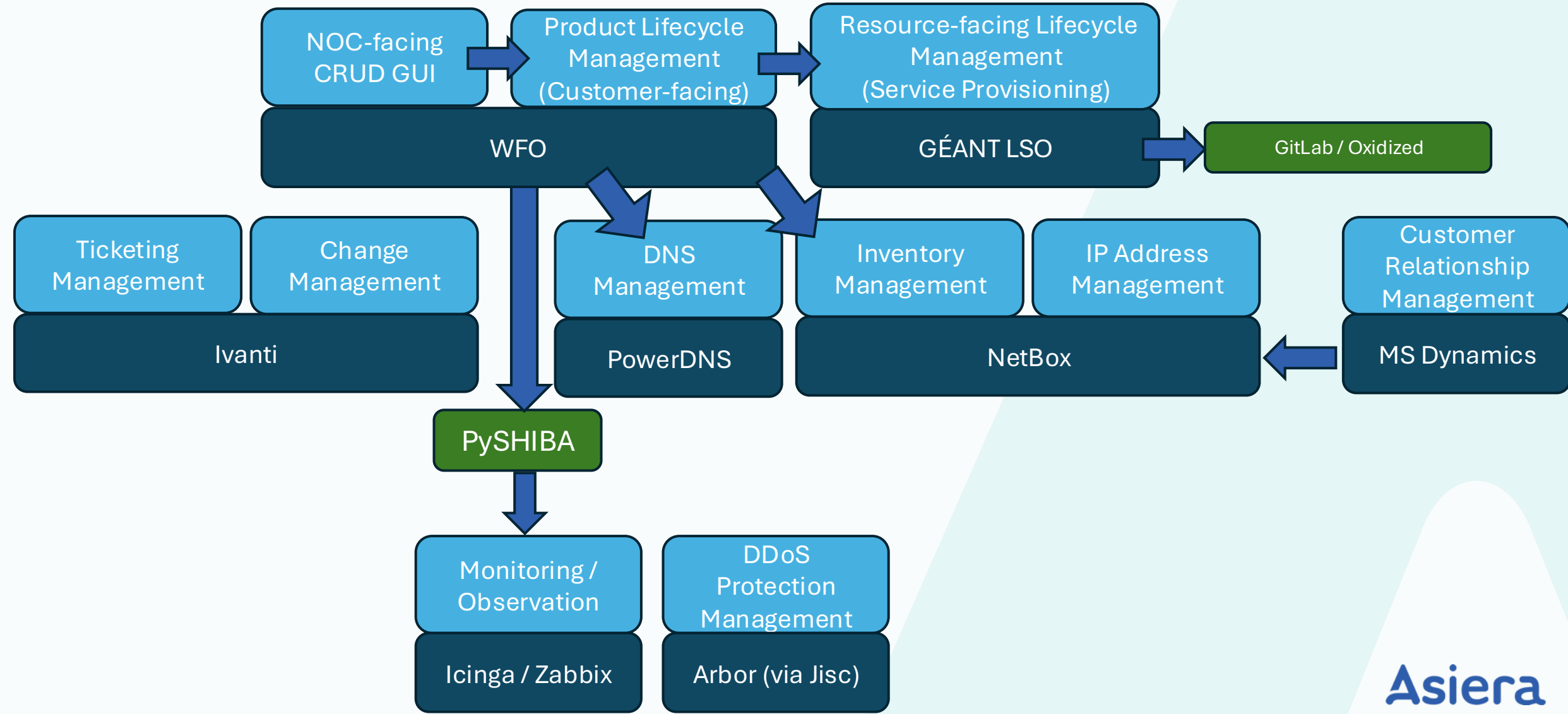
Asiera

Mick O'Donovan, Brian McArdle

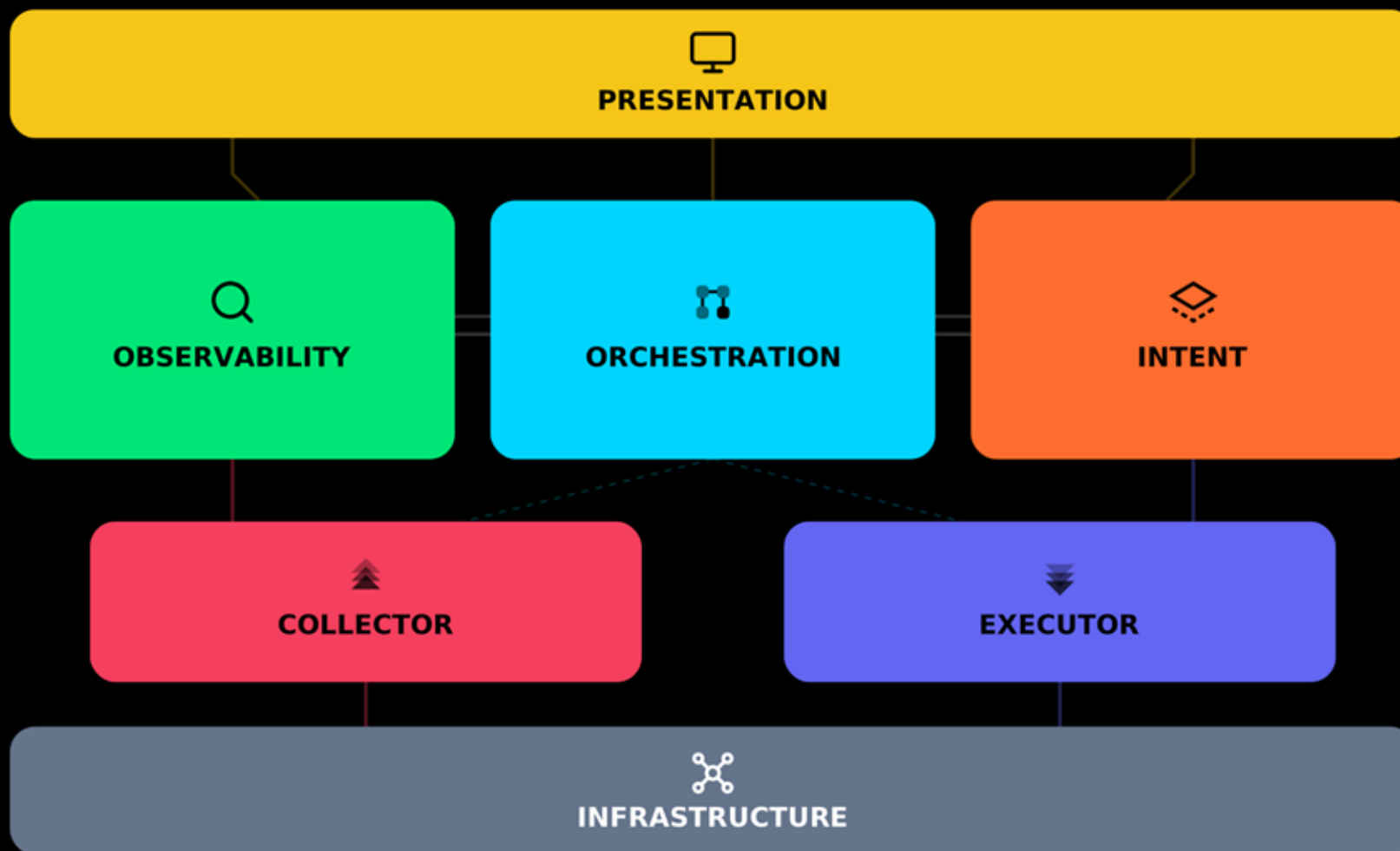
Sept 2026



Asiera OSS/BSS 2026



NAF Automation Framework



Asiera Automation Framework

ZABBIX

 **Grafana**



 **kentik**

NFSEN

 **Akvorado**
v2026.8.1



**Workflow
Orchestrator**

 **netbox**

 **gNMIc**

SNMP

The LS logo consists of the letters "LS" in a bold, sans-serif font, followed by a stylized asterisk or star symbol.

Lightweight Service Orchestrator

HPE

HPE JUNIPER
networking



Versions (May 2026 TNC -> Sept 2026)

■ WFO

- UI 7.2.0
- Orchestrator Core 4.4.2
- Scheduler
- Redis
- Celery

■ NetBox

- 4.5.8
- Extra workers

■ LSO

- 2.3.0

■ WFO

- UI 8.8.2
- Orchestrator Core 5.0.3
- Scheduler
- Redis
- Celery

■ NetBox

- 4.5.8
- Extra workers

■ LSO

- 2.4.6

Asiera

Advancing Services for Irish Education, Research & Academia

Next steps

- Implement RBAC
- Major upgrade of LSO
- Other minor upgrades
- Move everything to Kubernetes



Virtual Circuits

+ Add

⬆ Import

⬇ Export

Results1028

Filters

Quick search



⚙ Configure Table



☐	Circuit ID	Provider	Account	Provider Network	Type	Status	Tenant	Terminations	Description	Billable	Tags	Created ^	⌵
☐	tus-thurles.tus-limerick1.2ca99a73	HEAnet	—	Client or Internal Layer 2 Services	L2VPN Port-to-Port	Active	Technological University of the Shannon: Midlands Mid-West	2	edge1-tus-thurles1 et-0/0/2.0 <-> dist1-lit1 et-0/0/8.0	✓	—	2026-06-03 12:44:37	⌵
☐	tus-thurles.ipt.e9f39c7e	HEAnet	—	Client IP Transit Services	IP Transit VRRP	Active	Technological University of the Shannon: Midlands Mid-West	2	tus-thurles.ipt.e9f39c7e :: IPTVRRP 1000 Mbit/s :: edge1-tus-thurles1 et-0/0/0.12-edge2-tus-thurles1 et-0/0/0.12	✓	—	2026-06-02 14:52:43	⌵
☐	GN-INTERAGENCY.I3vpn.7a8d8c1e	HEAnet	—	L3 VPN Services	L3VPN	Active	GOVERNMENT NETWORKS	3	—	—	—	2026-06-02 08:47:55	⌵
☐	GN-MANAGEDACCESS.I3vpn.f1d80690	HEAnet	—	L3 VPN Services	L3VPN	Active	GOVERNMENT NETWORKS	4	—	—	—	2026-06-02 08:47:47	⌵
☐	GN-MULTIMEDIA.I3vpn.7d5308c8	HEAnet	—	L3 VPN Services	L3VPN	Active	GOVERNMENT NETWORKS	4	—	—	—	2026-06-02 08:47:34	⌵
☐	GN-SERVICES.I3vpn.e5bf9223	HEAnet	—	L3 VPN Services	L3VPN	Active	GOVERNMENT NETWORKS	4	—	—	—	2026-06-02 08:47:00	⌵
☐	tus-thurles.tus-thurles.c77077eb	HEAnet	—	Core Links	Core Link	Active	HEAnet	2	edge1-tus-thurles1 ge-0/0/3.0 <-> edge2-tus-thurles1 et-0/0/15.0	—	—	2026-05-29 11:00:38	⌵
☐	cwt.tus-thurles.0ae9cdb0	HEAnet	—	Core Links	Core Link	Active	HEAnet	2	edge3-cwt xe-0/0/41.0 <-> edge2-tus-thurles1 et-0/0/18.0	—	—	2026-05-29 10:40:59	⌵
☐	revenue1.ipt.28392656	HEAnet	—	Client IP Transit Services	IP Transit VRRP	Active	Met Éireann	2	revenue1.ipt.28392656 :: IPTVRRP 0 Mbit/s :: edge2-revenue ae1.1207-edge3-revenue1 ae1.1207	—	—	2026-05-27 09:58:31	⌵
☐	revenue1.ipt.db98c99f	HEAnet	—	Client IP Transit Services	IP Transit VRRP	Active	Met Éireann	2	revenue1.ipt.db98c99f :: IPTVRRP 0 Mbit/s :: edge2-revenue ae1.1205-edge3-revenue1 ae1.1205	—	—	2026-05-27 09:46:49	⌵
☐	revenue1.ipt.79df63bf	HEAnet	—	Client IP Transit Services	IP Transit VRRP	Decommissioned	Met Éireann	0	revenue1.ipt.79df63bf :: IPTVRRP 0 Mbit/s :: edge2-revenue ae1.1205-edge3-revenue1 ae1.1205	—	—	2026-05-27 09:42:20	⌵
☐	uofg-shannon.uofg1.7408300a	HEAnet	—	Client or Internal Layer 2 Services	L2VPN Port-to-Port	Active	University of Galway	2	edge1-uofg-shannon1 et-0/0/0.0 <-> edge1-gal1 xe-0/0/15.0	—	—	2026-05-25 11:29:40	⌵
☐	revenue1.ipt.39f95ee7	HEAnet	—	Client IP Transit Services	IP Transit Static	Active	Irish Prison Service	1	revenue1.ipt.39f95ee7 :: IPTStatic 1000 Mbit/s :: edge2-revenue et-0/0/9.0	—	—	2026-05-19 15:39:44	⌵
☐	setu-stkierans.wat-car1.ade1feba	HEAnet	—	Client or Internal Layer 2 Services	L2VPN Port-to-Port	Active	—	2	—	—	—	2026-05-18 15:49:14	⌵
☐	ucc-bio1.ucc-victoriamills1.c0ccf96d	HEAnet	—	Client or Internal Layer 2 Services	L2VPN Port-to-Port	Active	University College Cork	2	edge1-ucc-bio1 ge-0/0/0.0 <-> edge1-ucc-victoriamills1 ge-0/1/0.0	—	—	2026-05-14 10:46:34	⌵
☐	pw.pw.c9f7752e	HEAnet	—	Core Links	Core Link	Active	HEAnet	2	edge3-pw ae2.0 <-> edge4-pw ae2.0	—	—	2026-05-13 20:28:28	⌵
☐	pw.pw.4d1fb351	HEAnet	—	Core Links	Core Link	Active	HEAnet	2	core1-pw ae1.0 <-> edge3-pw ae1.0	—	—	2026-05-13 19:33:55	⌵
☐	tus-athlone2.pw.4cb503ae	HEAnet	—	Client or Internal Layer 2	L2VPN Port-to-Port	Active	Schools	2	tus-athlone2.pw.4cb503ae :: L2VPNPP 1000 Mbit/s	—	—	2026-05-13	⌵

Virtual Circuits

Results 1028 Filters

Quick search

▼

☐	CIRCUIT ID ▼	✖ PROVIDER	ACCOUNT	PROVIDER NETWORK	TYPE	STATUS	TENANT	TERMINATIONS
☐	dun1.ipt.b392c3de	HEAnet	—	Client IP Transit Services	IP Transit Commodity	Active	Dundalk Institute of Technology	2
☐	dun1.ipt.df2fd4d0	HEAnet	—	Client IP Transit Services	IP Transit Static	Active	HEAnet	1
☐	dun1.ipt.ee5c0b78	HEAnet	—	Client IP Transit Services	IP Transit VRRP	Decommissioned	Dundalk Institute of Technology	0
☐	dun1.pw.1df0ab46	HEAnet	—	Client or Internal Layer 2 Services	L2VPN Port-to-Port	Active	Schools	2
☐	dun1.pw.526cc215	HEAnet	—	Client or Internal Layer 2 Services	L2VPN Port-to-Port	Active	Schools	2
☐	dun1.pw.e827a117	HEAnet	—	Client or Internal Layer 2 Services	L2VPN Port-to-Port	Active	Schools	2
☐	dun1.pw.f9616b27	HEAnet	—	Client or Internal Layer 2 Services	L2VPN Port-to-Port	Active	Schools	2
☐	dun2.ipt.68d75b88	HEAnet	—	Client IP Transit Services	IP Transit Static	Decommissioned	HEAnet	0
☐	dun2.ipt.fa3563cd	HEAnet	—	Client IP Transit Services	IP Transit Static	Active	HEAnet	1

CPER

CPE replacement project
(over 180 routers at over 175
different locations)



CPE Replacement Project – 180+ routers

ACX2200



ACX7020

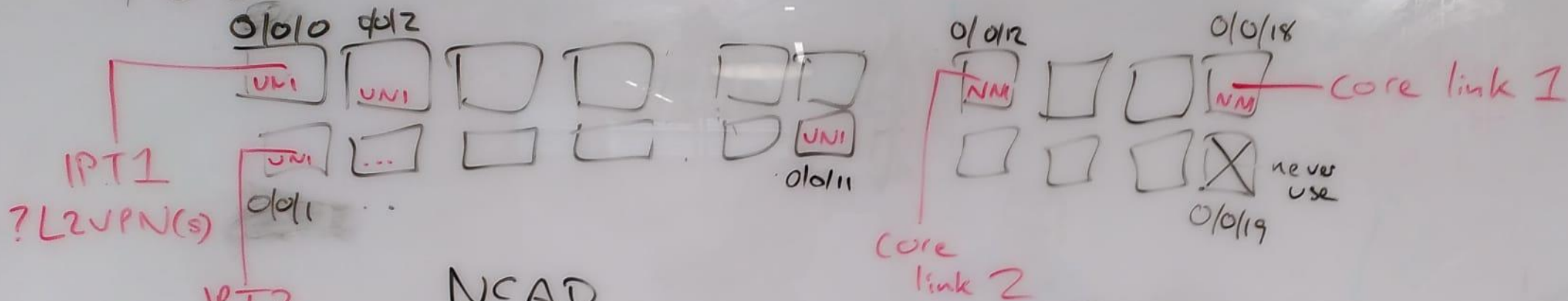


16 x SFP+ 1/10G
4 x SFP28 1/10/25G

CPER overview

- CPER – replacing equipment at ~175 sites around the country
 - - different ports between old and new devices
- Outside contractor doing the work
- Garwin Liu - automated production of microplans
 - needed predicted config
 - same services, different ports

7020



NCAD

ge-0/0/0 0/0/0 → 0/0/0
 0/1/0 0/0/1 → 0/0/1
 0/1/3 0/0/1
 0/2/1 0/0/2

Core link

- 1 → goes to core network always
→ check if desc for core device name
- 2 → goes to core network
OR other device on same site
→ site name is in if desc

ΣN-KP1

ge 0/0/0 0 IPT 0/0/0
 0/1/0 72 L2VPN 0/0/1
 0/1/1 937 L2VPN 0/0/2
 xe - 0/3/0 0 NNI - 0/0/18
 0/3/1 3300 L2VPN 0/0/3

↓
 different PPS on source
 ⇒ different PPS on dest

Workflow 1

Generate expected current config

```
for a node sub
  foreach port on node
    foreach service on port
      generate config
```

=> config for entire current node

(This will be useful for RMA workflow)

Workflow 2

Generate predicted config for CPER

```
for a node sub
  foreach port on node
    predict $newport based on mapping
    foreach service on port
      generate config using $newport
```

=> config for entire future node

```
def generate_config() -> StepList:
    return (
        begin
        >> lso.anonymous_indifferent_lso_interaction(call_node_config_playbook)
        >> display_node_base_config
        >> display_node_local_export_config
        >> display_node_model_specific_config
        >> display_port_config
        >> display_core_port_config
        >> display_ipt_config
        >> display_l2vpn_config
        >> display_l3vpn_config
        >> display_core_link_config
        >> print_generated_config
    )
```

```
def generate_config_cper() -> StepList:
    return (
        begin
        >> find_associated_ports_and_subscriptions
        >> classify_ports
        >> check_core_ports
        >> assign_core_ports
        >> assign_service_ports
        >> lso.anonymous_indifferent_lso_interaction(call_node_config_playbook)
        >> display_node_base_config
        >> display_node_local_export_config
        >> display_node_model_specific_config
        >> display_port_config
        >> display_core_port_config
        >> display_ipt_config
        >> display_l2vpn_config
        >> display_core_link_config
        >> display_l3vpn_config
        >> download_config
        # >> send_mail_with_textfile
    )
```

```

class L2vpnPPPProvisioning(L2vpnPPInactive, lifecycle=[SubscriptionLifecycle.PROVISIONING]):
    def ansible_variables(self, temp_node: str="", temp_port: str="") -> dict:
        port_A = self.virtual_circuit.sap_a.port
        device_A = netbox.get_device(name=self.virtual_circuit.sap_a.port.node.node_name)
        port_B = self.virtual_circuit.sap_b.port
        device_B = netbox.get_device(name=self.virtual_circuit.sap_b.port.node.node_name)
        port_A_name = port_A.port_name
        port_B_name = port_B.port_name

        # check if we are to generate config for a future interface
        # and which end it is
        if temp_node == self.virtual_circuit.sap_a.port.node.node_name:
            port_A_name = temp_port
            device_A = netbox.get_device(name=temp_node)
        if temp_node == self.virtual_circuit.sap_b.port.node.node_name:
            port_B_name = temp_port
            device_B = netbox.get_device(name=temp_node)

        device_A_primary_ip_without_mask = str(ipaddress.ip_interface(device_A.primary_ip.address).ip)
        device_B_primary_ip_without_mask = str(ipaddress.ip_interface(device_B.primary_ip.address).ip)

        return ({
            "vc_id": self.virtual_circuit.vc_id,
            "interface_description": f"(UN) {self.title} vpn/l2vpns/{self.virtual_circuit.ims_id}",
            # mtu is set the same on both SAPs
            "mtu": self.virtual_circuit.saps[0].mtu,
            "SiteA": f"{device_A.name}.nn.hea.net",
            "interfaceA": port_A_name,
            "p2p_endpointB": device_A_primary_ip_without_mask,
            "SiteB": f"{device_B.name}.nn.hea.net",
            "interfaceB": port_B_name,
            "p2p_endpointA": device_B_primary_ip_without_mask,
        })

```

Adding a new API endpoint

1. Add new endpoint

If not already existing, add a new folder tree - `api/v1/endpoints`.

In that folder, add your `endpoint.py`.

For example `api/v1/endpoints/download_tenant.py`:

```
from uuid import UUID
import requests
from fastapi import APIRouter, HTTPException, Request
from fastapi.responses import PlainTextResponse

router = APIRouter(tags=["download"])

@router.get("/download_tenant/{node_name}")
def download_tenant(node_name: str, request: Request):
    device = netbox.get_device(name = node_name)

    if not device:
        raise HTTPException(status_code=404, detail=f"No {node_name} found")

    return device.tenant
```

2. Add route

In `main.py`:

```
from api.v1.endpoints.download_tenant import router as download_tenant_router
# Include your custom endpoints under /api/v1
app.include_router(download_tenant_router, prefix="/api/v1")
```

3. Update auth exceptions

Our API has IP restrictions but no capability for authentication using tokens.

The new endpoint must be added to the exceptions in `auth/oidc.py`:

```
if request.url.path.startswith("api/v1/download_tenant/"):
    return True
```

Adding an endpoint that pulls data from a workflow

This is a complication version of step 1 above. It assumes a workflow of `wf_id` has been run, that within it generates a state variable called `generated_config`.

The `node_name` in the URI is purely for the file name. The device name could also be extracted from the workflow state if so desired.

We can access this endpoint via `wfo-api.heanet.ie/api/v1/download/{wf_id}/{subscription.node.node_name}/download_cper_config`

```
router = APIRouter(tags=["download"])

@router.get("/download/{wf_id}/{node_name}/download_cper_config")
def download_cper_config(wf_id: UUID, node_name: str, request: Request):
    # get details of workflow from API call
    base_url = str(request.base_url).rstrip("/") # e.g. https://wfo.heanet.ie
    url = f"{base_url}/api/processes/{wf_id}"
    print(f"Fetching process details from {url}")
    process_details = requests.get(url)
    print(f"Process details: {process_details}")

    if not process_details:
        raise HTTPException(status_code=404, detail="Workflow not found")
    # transform to json
    process_details = process_details.json()

    # get config from details
    if "current_state" not in process_details:
        raise HTTPException(status_code=404, detail="No current_state in workflow state")
    if "generated_config" not in process_details['current_state']:
        raise HTTPException(status_code=404, detail="No generated_config in workflow state")

    content = process_details['current_state']['generated_config']

    # push file to user
    return PlainTextResponse(
        content,
        media_type="text/plain",
        headers={"Content-Disposition": f'attachment; filename="{node_name}_cper_config.txt"'})
```



Download config as text file
success - 9/22/2026, 9:17:55 AM

```
1 {
2   | "download_url": "http://wfo-api.heanet.ie/api/v1/download/32b63228-614a-4a2c-a692-b7b5606aecd5/edge1-heanet-nd2/download_cper_config"
3 }
```

Duration

00:00:00

Migration script – 1/5

- 1. Modify WFO node sub**
2. Validate sync state
3. Discover ports and services
4. Create new Netbox interfaces
5. Build and confirm port mapping
6. Create new WFO port subs
7. Modify WFO service subs
8. Terminate old WFO port subs
9. Delete old Netbox interfaces
10. Rename Netbox management interface
11. Refresh SNMP indexes via WFO task

orchestrator@545aaaaee7092:~/etc/wfo-scripts\$ python cper-migration.py

Enter device name: edge1-ucc-beaufort
Enter new asset tag: 36657

← User Input

Checking NetBox for device type 'ACX7020-AC'...

Found device type: ACX7020-AC

--- Device Information ---

Name: edge1-ucc-beaufort
New asset Tag: 36657
New type: ACX7020-AC

Use these details? (y/n): y

← User Input

User Input (optional)

Do you wish to continue in dry run mode? Enter 'REAL' to perform real migration, anything else to continue in dry mode:

Device information complete! Attempting to find WFO subscription to update...
Found matching subscription with ID 2382573b-4f33-4c6f-80e7-63497efb2be5

Modifying edge1-ucc-beaufort with new asset tag and device type in WFO/NetBox

[DRY-RUN] Would POST to https://OUR_WFO/api/processes/modify_node with payload:
[{'subscription_id': '2382573b-4f33-4c6f-80e7-63497efb2be5'}, {'asset_tag': '36657', 'type_id': '30', 'node_status': ...
{'id': 'dry-run-id', 'last_status': 'completed'}]

Device updated in WFO and NetBox!

Migration script – 2/5

1. Modify WFO node sub
2. **Validate sync state**
3. **Discover ports and services**
4. **Create new Netbox interfaces**
5. **Build and confirm port mapping**
6. Create new WFO port subs
7. Modify WFO service subs
8. Terminate old WFO port subs
9. Delete old Netbox interfaces
10. Rename Netbox management interface
11. Refresh SNMP indexes via WFO task

Check for objects in non-valid state...
Finding ports and services on this device...

Found active core / LAG port on device: `ge-0/2/1`
Found active CoreLink subscription on device: `edge1-ucc-beaufort`
Found active port/LAG on device: `ge-0/1/0`

Found active L2VPNPP subscription on device: `cork1.ucc-beaufort.2257834d` :: `L2VPNPP 1000 Mbit/s`

All ports and services are valid!

Attempting to create ACX7020-AC interfaces on `edge1-ucc-beaufort` in NetBox...

[DRY-RUN] Would create port `et-0/0/0`
[DRY-RUN] Would create port `et-0/0/1`
[DRY-RUN] Would create port `et-0/0/2`
...
[DRY-RUN] Would create port `et-0/0/19`

Interfaces created in NetBox!

All ports and services scanned.

Classifying ports per service type...
Classifying port: `ge-0/2/1` with `1` services - classified as `CORE` port
Classifying port: `ge-0/1/0` with `1` services - classified as `L2VPN` port

Classification results: `1 CORE, 0 IPT, 1 L2VPN, 0 L3VPN`

Discovered 1 core ports to migrate

Found neighbour name `edge1-cork2` in `[NN]` `edge1-cork2 xe-0/0/41 (1550nm)`

Checking neighbour_type `ACX5448-M` in `['ACX5448-M', 'ACX7024-AC']`
`is_core_facing?` `True`

Comparing current SG `UCC Beaufort Research Lab` with neighbour SG `University College Cork Campus`
`is_same_site?` `False`

Assigned single **CORE** link `ge-0/2/1 → et-0/0/18` **(Rule 1)**

Discovered 1 client ports to migrate

Assigned **L2VPN** port `ge-0/1/0 → et-0/0/0`

All service ports re-mapped

Mapping of ports complete!

- all services on `ge-0/2/1` will be moved to `et-0/0/18`
- all services on `ge-0/1/0` will be moved to `et-0/0/0`

Is this mapping correct, and does it match the generated config? (y/n): `y` ← User Input

Migration script – 4/5

1. Modify WFO node sub
2. Validate sync state
3. Discover ports and services
4. Create new Netbox interfaces
5. Build and confirm port mapping
- 6. Create new WFO port subs**
- 7. Modify WFO service subs**
- 8. Terminate old WFO port subs**
9. Delete old Netbox interfaces
10. Rename Netbox management interface
11. Refresh SNMP indexes via WFO task

```
[DRY-RUN] would create new port subscription for et-0/0/18
[DRY-RUN] find sub for new port
[DRY-RUN] setting UUID for new port to 12345678
```

Port subscription created

Migrating services from ge-0/2/1 to et-0/0/18.

Modifying CORE service cork2.ucc-beaufort.f1b186e8 edge1-cork2 xe-0/0/41 <-> ge-0/2/1 edge1-ucc-beaufort to use new port...

Sending new data to WFO to https://OUR_WFO/api/processes/modify_core_link: ...

Terminate old port
Terminating ge-0/2/1 on edge1-ucc-beaufort in WFO

...
Port subscription created

Migrating services from ge-0/1/0 to et-0/0/0.

Modifying L2VPN service cork1.ucc-beaufort.2257834d :: L2VPNPP 1000 Mbit/s to use new port...

Sending new data to WFO to https://OUR_WFO/api/processes/modify_l2vpn_pp: ...

Terminate old port
Terminating ge-0/1/0 on edge1-ucc-beaufort in WFO

...
Port terminated

Migration script – 5/5

1. Modify WFO node sub
2. Validate sync state
3. Discover ports and services
4. Create new Netbox interfaces
5. Build and confirm port mapping
6. Create new WFO port subs
7. Modify WFO service subs
8. Terminate old WFO port subs
- 9. Delete old Netbox interfaces**
- 10. Rename Netbox management interface**
- 11. Refresh SNMP indexes via WFO task**

Delete old interfaces from NetBox

Deleting old interface **ge-0/0/0**
Deleting old interface **ge-0/0/1**
...
Deleting old interface **xe-0/3/1**

All old interfaces deleted

Renaming local management interface

*[DRY-RUN] Would update NetBox interface: **re0:mgmt-0***

Renamed local management interface

Refreshing SNMP i/f indexes
Refreshing SNMP indexes for all ports on **edge1-ucc-beaufort** in WFO

[DRY-RUN] Would POST to https://OUR_WFO/api/processes/update_snmp_if_indexes with payload: ...
[DRY-RUN] Skipping workflow polling – assuming completed

SNMP ifIndexes refreshed

Icinga and Cacti monitoring regenerated

Migration completed successfully

CPER – the end result

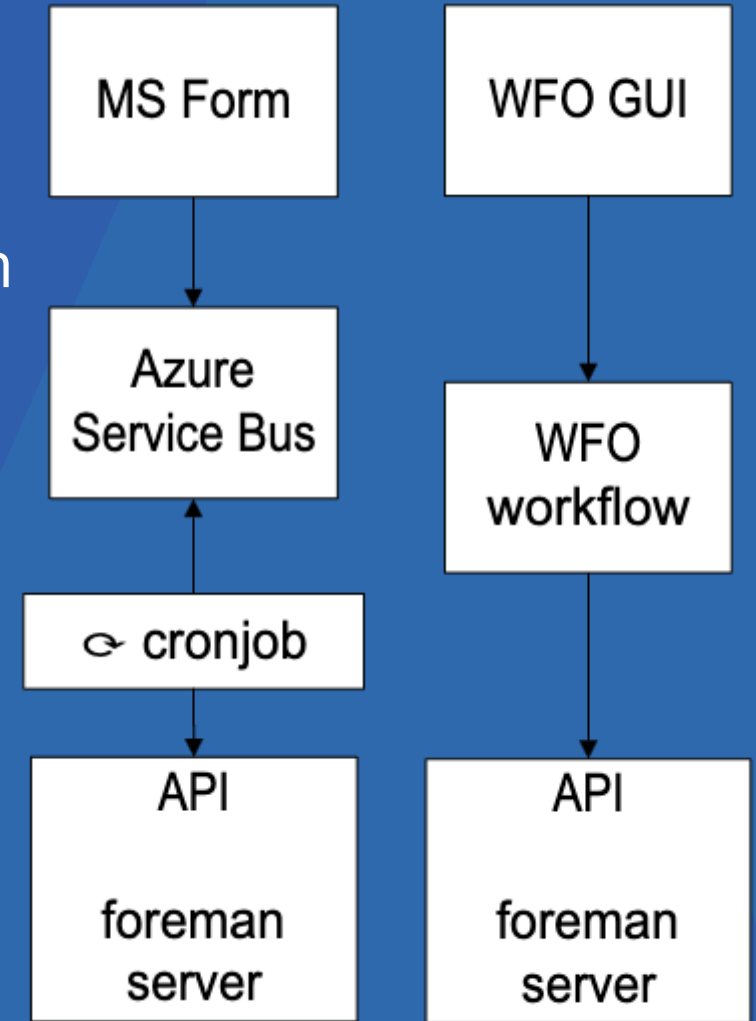
1. Asiera Check Subscription in Sync
2. Asiera Configuration Generation : Modify Workflow
3. Partner goes to site with new appliance
4. Asiera Day of Migration – Run The Migration Script
5. Migration Complete

3 migrations per day vs. **2 migrations per week!**

As of the end of September 2026 – **178 migrations completed!**

Inventory management

- Server inventory for fleet management: Foreman
 - Lots of automation
 - Few edge cases of devices that are manually added
- Previously: Microsoft workflow and local script
 - Highly resilient
- Then: Microsoft changed licensing so €€€
- Now: replace with WFO task
 - Not highly resilient, but simple to retry on failure



Testing

- Run each workflow
- Not reliable to run on live devices
- Credit to Mark of SURF – skip_config
- Fixtures for data – note use of strings rather than IDs

Auto Negotiation

☐ Auto Negotiation

Lldp

☐ Lldp

Tick to skip LSO/Ansible calls

Skip Config

☐ Skip Config

Previous

Cancel

```
@create_workflow(initial_input_form=initial_input_form_generator)
```

```
def create_port() -> StepList:
```

```
    # if skip_config is true, this returns conditional to skip LSO steps
```

```
    def skip_config() -> Callable:
```

```
        def check_skip_config(state: dict) -> bool:
```

```
            # print(f"skip_config() called, returning {bool(state['skip_config'] is False)}")
```

```
            return state['skip_config'] is False
```

```
        return conditional(check_skip_config)
```

```
    return (
```

```
        begin
```

```
        >> construct_port_model
```

```
        >> store_process_subscription()
```

```
        >> enable_port
```

```
        >> lso.anonymous_indifferent_lso_interaction(call_ansible_generate_playbook_live)
```

```
        >> skip_config()(lso.indifferent_pp_interaction(call_ansible_playbook_dryrun))
```

```
        >> print_gitlab_link
```

```
        >> skip_config()(lso.indifferent_pp_interaction(call_ansible_playbook_live))
```

```
        >> print_gitlab_link
```

```
        >> update_port_in_ims
```

```
        >> create_port_in_pyshiba
```

```
)
```

Testing

```
! node.yml ×
tests > fixtures > ! node.yml > { } node_one
1  node_one:
2    product_name: "node Juniper"
3    name: "edge-test-ci-node-mx240"
4    description: "Test node for CI pipeline"
5    type: "Juniper MX240"
6    role: "Edge Router"
7    site: "North Dock Two"
8    node_status: "active"
9  node_two:
10   product_name: "node Juniper"
11   name: "dist-test-ci-node-mx104"
12   description: "Test node for CI pipeline"
13   type: "Juniper MX104-PREMIUM"
14   role: "Dist Router"
15   site: "North Dock Two"
16   node_status: "active"
17  node_three:
18   product_name: "node Juniper"
19   name: "edge-test-ci-node-acx2200"
20   description: "Test node for CI pipeline"
21   type: "Juniper ACX2200-AC"
22   role: "Edge Router"
23   site: "North Dock Two"
24   node_status: "active"
25  node_four:
26   product_name: "node Juniper"
27   name: "core-test-ci-node-cwt-acx2200"
```

Testing

```
# start new creation workflow
try:
    print(f"Posting full data for {l2vpn['device_a']} {l2vpn['port_a']} / {l2vpn['device_b']} {l2vpn['port_b']} to WFO")
    response = wfo.post(product_url, json=request_data)
except Exception as err:
    print(f"Posting data to {product_url} failed: {err} ({request_data})")

print(response.text)
response = response.json()

if 'id' not in response:
    exit(f"Workflow not created: {response}")

# print(response)
process_id = response['id']

# get status of workflow
try:
    process_state = wfo.get(f"{wfo_url}/processes/{process_id}").json()
except Exception as err:
    print(f"Can not fetch process_state from {wfo_url}/processes/{process_id}: {err}")

# print(f"Current process_state: {process_state['last_status']}")

# wait while workflow proceeds
while process_state['last_status'] != 'completed' and process_state['last_status'] != 'failed':
    print(process_state['last_status'])
    sleep(5)
    process_state = wfo.get(f"{wfo_url}/processes/{process_id}").json()

# if workflow completes, exit with success
if process_state['last_status'] == "completed":
    print(f"Success: {process_state['last_status']}")
else:
    # else, exit with non-zero failure
```

Testing

- Run automatically via Gitlab CI
- Wait for workflow to be “completed” or “failed”
- Return status
- Other staging services can be unreliable
- But still very useful to prevent regression bugs

Testing - success

```
182 11:58:21 $ python tests/functional/l2vpn_pp_create_test.py
183 11:59:10 Posting product_id, expecting tenant selection page
184 11:59:10 Posting partial data, expecting port selection page
185 11:59:10 "type":"FormNotCompleteError","detail":"'${defs': {'PortsEnum': {'enum': ['7ca91829-d58c-442d-90c2-73a1b9e30c72',
'ff73b77e-1de6-422e-94f7-8ca82e4443fe', 'ac960e0b-0a4a-4801-8f7c-38f429bdab6e', '6fd5093a-4b5c-4158-9a46-4ccda5b1
5e5d', '64aff4bd-0d57-...
186 11:59:10 Posting full data for dist-test-ci-node-mx104 xe-2/0/0 / edge-test-ci-node-acx2200 ge-0/0/0 to WF0
187 11:59:10 {"id":"bcd31131-692b-4440-ba2e-270e84f162e7"}
188 11:59:10 awaiting_callback
189 11:59:10 awaiting_callback
190 11:59:10 awaiting_callback
191 11:59:10 awaiting_callback
192 11:59:10 running
193 11:59:10 running
194 11:59:10 running
195 11:59:10 Success: completed
196 11:59:10 All workflows completed
197 11:59:10 $ python tests/functional/l2vpn_pp_terminate_test.py
198 11:59:19 Searching for correct subscription to terminate.....Found matching VC ID...
199 11:59:19 Found matching VC ID and matching ports: heanet-nd2.heanet-nd2.ff3ecb1c :: L2VPNPP
200 11:59:19 Posting data for heanet-nd2.heanet-nd2.ff3ecb1c :: L2VPNPP to WF0
201 11:59:19 {"id":"dcde6c70-2ee3-4a9a-8ede-6033396a8e28"}
202 11:59:19 Success: completed
203 11:59:19 All workflows completed
```

Duration: 28 minutes 20 seconds

Finished: Sep 18, 2026, 13:20

Queued: 1 minute 16 seconds

Timeout: 1h (from project) ?

Runner: #78 (KyugsL-k)
gitlabrunner-prod-pw-07

Source: Push

Tags: wfo-lab-access

Commit dc954bed

Merge branch '183-ipt-dns' into dev

Pipeline #687387 ! Warning for de
v

functional-testing

Related jobs

✓ cper_functional_tests

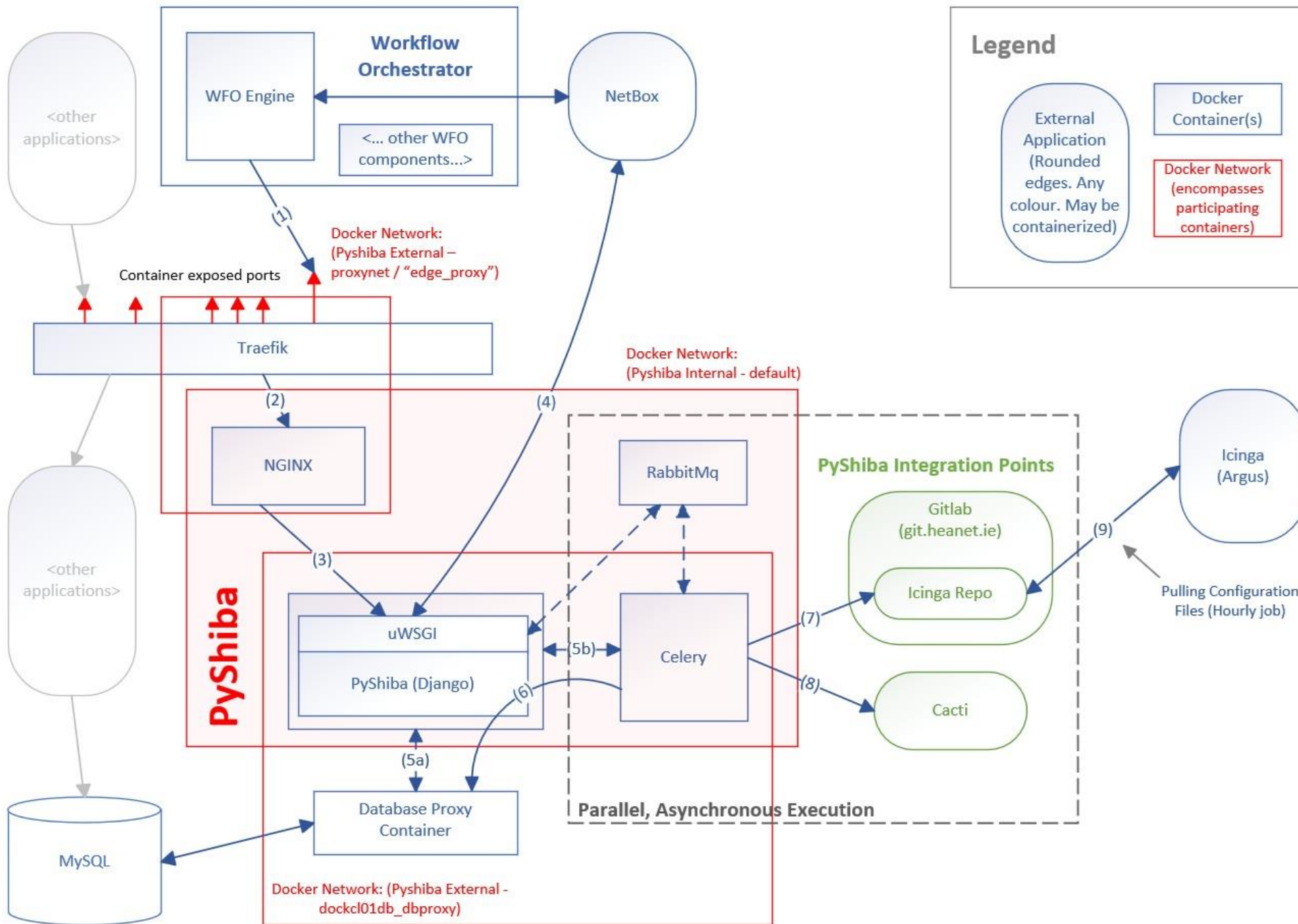
→ ✓ functional_tests

✓ python-validation

Testing - failure

```
54 11:26:02 $ python tests/functional/node_create_test.py
55 11:26:51 Posting product_id, expecting details selection page
56 11:26:51 Posting full data for edge-test-ci-node-mx240 to WFO
57 11:26:51 "id":"63b06045-7eff-4e0d-87d9-3b4ea052e74d"}...
58 11:26:51 Status: failed
59 11:26:51 Reason: HTTPSConnectionPool(host='wfo-lab-netbox.heatnet.ie', port=443): Max retries exceeded with url: /api/tenancy/tenants/?name=Asiera&limit=0 (Caused by SSLError(SSL error(1, '[SSL] record layer failure (_ssl.c:2660)'))))
60 11:26:51 Node: {'product_name': 'node Juniper', 'name': 'edge-test-ci-node-mx240', 'description': 'Test node for CI pipeline', 'type': 'Juniper MX240', 'role': 'Edge Router', 'site': 'North Dock Two', 'node_status': 'active'}
61 11:26:51 Posting product_id, expecting details selection page
62 11:26:51 Posting full data for dist-test-ci-node-mx104 to WFO
63 11:26:51 "id":"c467fc52-3b54-4748-b4af-d23d7dcc062f"}...
64 11:26:51 Success: completed
65 11:26:51 Posting product_id, expecting details selection page
66 11:26:51 Posting full data for edge-test-ci-node-acx2200 to WFO
67 11:26:51 "id":"9ea81924-40d5-4952-960e-d8d7f5e79f62"}...
68 11:26:51 Success: completed
69 11:26:51 Posting product_id, expecting details selection page
70 11:26:51 Posting full data for core-test-ci-node-cwt-acx2200 to WFO
71 11:26:51 "id":"f180e5cc-c400-41ba-8d5f-c71046911e09"}...
72 11:26:51 Success: completed
73 11:26:51 All workflows completed
74 11:26:51 Running after_script
75 11:26:51 Running after script...
76 11:26:51 $ python tests/functional/core_port_terminate_test.py
77 11:26:52 Did not find matching sub
78 11:26:52 Did not find matching sub
79 11:26:52 Did not find matching sub
80 11:26:52 Did not find matching sub
81 11:26:52 All workflows completed
82 11:26:52 $ python tests/functional/port_terminate_test.py
83 11:26:54 Did not find matching sub
84 11:26:54 Did not find matching sub
```

00:33



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Thank you!
Any more questions?

